

BITS Pilani, Pilani Campus

Chemical Engineering Department

Chemical Engineering Lab – I

Experiment 10

Cross-Circulation Drying

Submitted by : _____
ID No. : _____
Section : P-1
Group : 2
Date of Experiment : 09/09/2026

1 Aim

Study of drying characteristics of porous solids under forced draft condition with cross flow of air.

Objectives

1. To determine the critical moisture content.
2. To calculate the total drying time.
3. To calculate the humidity of inlet and outlet air with respect to drying time.

2 Formulae Used

Quantity	Formula
Total (free) moisture content	$X_T = \frac{w_3 - w_2}{w_2 - w_1}$
Rate of drying	$R = -\frac{m_s}{A} \frac{dX}{dt}$
Total time of drying	$t_T = \frac{m_s}{AR_c} \left[(X_1 - X_c) + X_c \ln \frac{X_c}{X_2} \right]$
Saturation vapour pressure (Tetens)	$p_{ws}(T) = 0.6113 \exp\left(\frac{17.67 T}{T + 243.5}\right)$ [kPa, T in °C]
Carrier's equation (vapour pressure)	$p_w = p_{ws}(T_{wb}) - 0.00066 P (T_{db} - T_{wb})(1 + 0.00115 T_{wb})$
Humidity ratio	$Y = \frac{0.622 p_w}{P - p_w}$

where w_1 = weight of empty pan, w_2 = weight of (pan + dry solid), w_3 = weight of (pan + wet solid) at time t , m_s = mass of bone-dry solid, A = surface area of pan, X_1, X_2 = initial and final free-moisture content, X_c, R_c = free-moisture content and rate of drying at the critical point, P = atmospheric pressure, T_{db}, T_{wb} = dry- and wet-bulb temperatures.

3 Data / Specifications

Material	Brick particles (porous)
Surface area of pan, A	0.0245 m ²
Air velocity	4 m/s
Weight of empty pan, w_1	270 g
Weight of (pan + dry solid), w_2	520 g
Mass of bone-dry solid, $m_s = w_2 - w_1$	250 g = 0.250 kg
Atmospheric pressure, P	101.325 kPa (assumed standard)

Note on w_2 : the manual specifies w_2 as the weight of (pan + dry solid) measured at step 5, before wetting. This value was not separately recorded on the data sheet. Per the manual's own criterion in step 12 – that the pan-plus-solid weight at the end of a completed run should return to nearly the same value noted before wetting – the final steady reading, $T_{20} = 520$ g, is used as a proxy for w_2 . This gives $m_s = 250$ g, consistent with the manual's instruction to charge 'about 250 gm of brick particles' (Sec. 5, step 2), supporting the validity of this assumption.

4 Observations

Table 1: Raw readings – weight and inlet/outlet wet- and dry-bulb temperatures

S.No	Time (min)	w_3 (g)	Inlet (°C)		Outlet (°C)	
			T_{wb}	T_{db}	T_{wb}	T_{db}
T0	0	575	26.0	35.0	28.0	41.0
T1	2	573	29.0	40.0	29.0	42.0
T2	4	572	29.0	43.0	31.5	43.0
T3	6	571	31.0	45.0	32.0	46.0
T4	8	570	32.0	47.0	34.0	49.0
T5	10	568	32.5	48.0	35.0	50.0
T6	12	566	32.5	48.0	36.0	51.0
T7	14	565	32.5	49.0	37.0	52.0
T8	16	563	32.5	50.0	37.0	53.0
T9	18	561	33.0	50.0	38.0	54.0
T10	20	560	33.0	50.0	39.0	54.0
T11	25	555	33.0	51.0	39.0	53.0
T12	30	550	35.0	51.0	40.0	54.0
T13	35	546	35.0	52.0	40.0	54.0
T14	40	542	35.0	52.0	40.0	55.0
T15	45	538	35.0	53.0	40.0	55.0
T16	50	534	35.0	53.0	40.0	55.0
T17	55	530	35.0	53.0	41.0	55.0
T18	60	525	35.0	53.0	41.0	56.0*
T19	65	522	35.0	53.0	41.0	55.0
T20	70	520	35.0	53.0	41.0	55.0

*T18 outlet dry-bulb reading was overwritten on the data sheet (digit corrected); read and used as 56°C. Time interval was 2 min for T0–T10 and 5 min for T11–T20, as recorded.

5 Sample Calculations

Illustrated for $t = 10$ min (T5): $w_3 = 568$ g, inlet $T_{wb} = 32.5^\circ\text{C}$, $T_{db} = 48^\circ\text{C}$.

(a) Total moisture content

$$X_T = \frac{w_3 - w_2}{w_2 - w_1} = \frac{568 - 520}{520 - 270} = \frac{48}{250} = 0.192 \text{ kg moisture/kg dry solid}$$

(b) Rate of drying

Because the raw weight readings are quantised to the nearest gram, a point-to-point finite difference of X_T is dominated by rounding noise. A cubic smoothing spline was fitted through $X_T(t)$ and differentiated analytically to obtain a smoothed dX/dt at each time:

$$\left. \frac{dX}{dt} \right|_{t=10} = -0.00309 \text{ min}^{-1}$$

$$R = -\frac{m_s}{A} \frac{dX}{dt} = \frac{0.250}{0.0245} \times 0.00309 = 0.0316 \text{ kg}/(\text{m}^2 \cdot \text{min})$$

(c) Humidity (Carrier's equation)

Saturation vapour pressure at $T_{wb} = 32.5^\circ\text{C}$:

$$p_{ws} = 0.6113 \exp\left(\frac{17.67 \times 32.5}{32.5 + 243.5}\right) = 4.897 \text{ kPa}$$

Vapour pressure of inlet air:

$$p_w = 4.897 - 0.00066 \times 101.325 \times (48 - 32.5)(1 + 0.00115 \times 32.5) = 3.821 \text{ kPa}$$

Humidity ratio:

$$Y = \frac{0.622 \times 3.821}{101.325 - 3.821} = 0.0244 \text{ kg water vapour/kg dry air}$$

6 Calculation Table

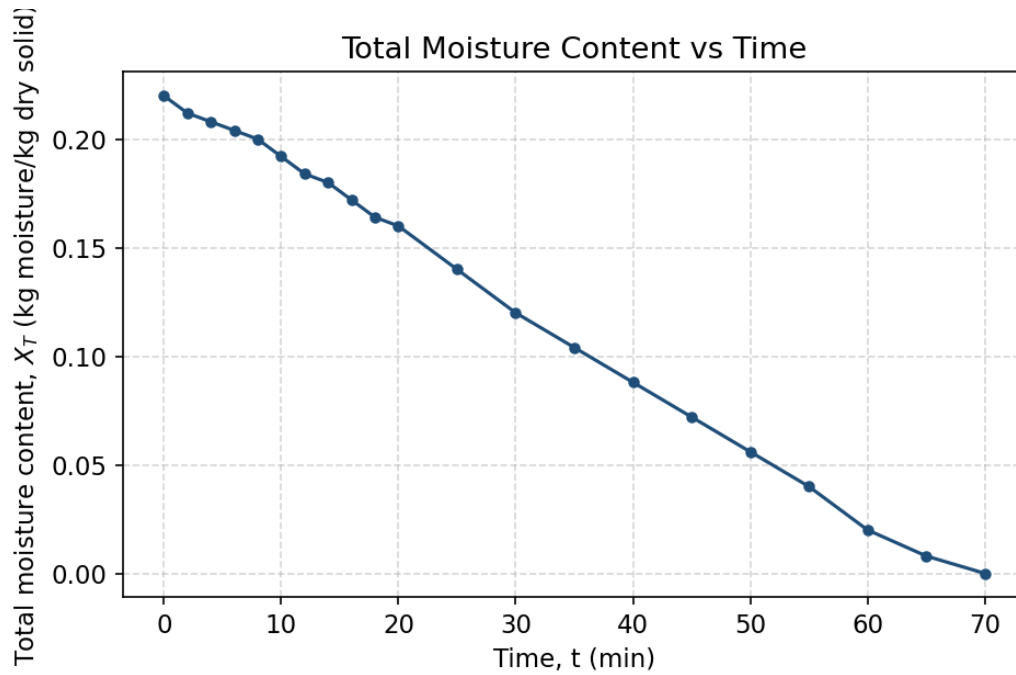
Table 2: Moisture content, rate of drying and humidity at every reading

S.No	t (min)	X_T (kg/kg)	R (kg/m ² .min)	Y_{in} (kg/kg)	Y_{out} (kg/kg)
T0	0	0.2200	0.0261	0.0173	0.0182
T1	2	0.2120	0.0273	0.0206	0.0197
T2	4	0.2080	0.0285	0.0192	0.0244
T3	6	0.2040	0.0296	0.0224	0.0242
T4	8	0.2000	0.0306	0.0237	0.0275
T5	10	0.1920	0.0316	0.0244	0.0295
T6	12	0.1840	0.0324	0.0244	0.0317
T7	14	0.1800	0.0332	0.0239	0.0339
T8	16	0.1720	0.0339	0.0235	0.0335
T9	18	0.1640	0.0345	0.0246	0.0359
T10	20	0.1600	0.0350	0.0246	0.0389
T11	25	0.1400	0.0360	0.0242	0.0394
T12	30	0.1200	0.0365	0.0290	0.0421
T13	35	0.1040	0.0364	0.0286	0.0421
T14	40	0.0880	0.0358	0.0286	0.0416
T15	45	0.0720	0.0348	0.0281	0.0416
T16	50	0.0560	0.0332	0.0281	0.0416
T17	55	0.0400	0.0346 [†]	0.0281	0.0449
T18	60	0.0200	0.0285	0.0281	0.0444
T19	65	0.0080	0.0254	0.0281	0.0449
T20	70	0.0000	0.0218	0.0281	0.0449

[†]Row highlighted: taken as the critical point (Sec. 7).

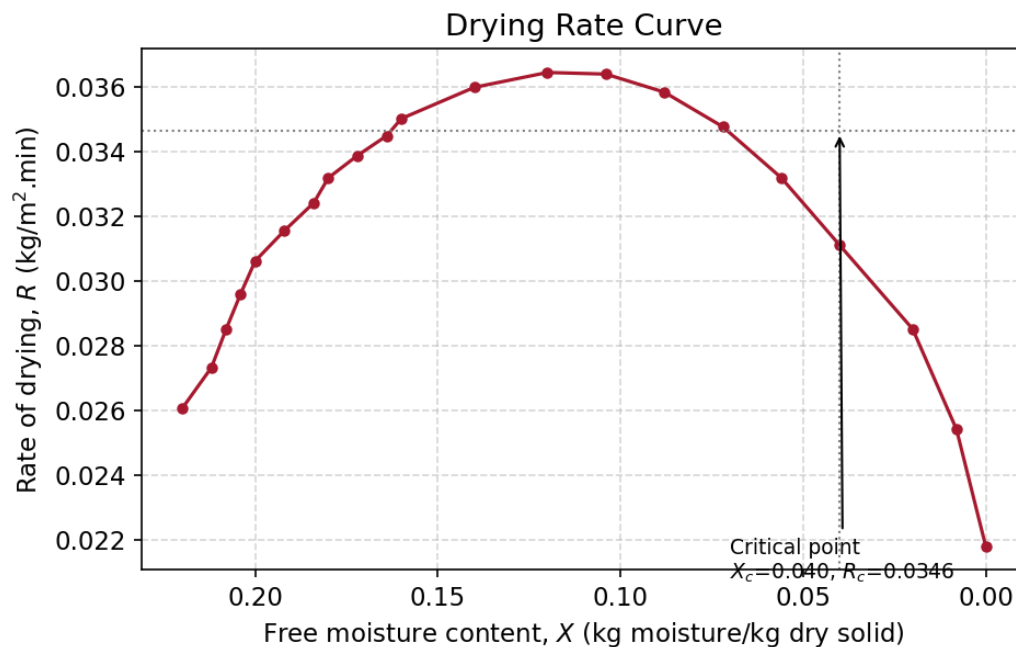
7 Results

7.1 Drying curve: total moisture content vs. time



X_T falls smoothly and almost monotonically from 0.220 to 0.000 kg moisture/kg dry solid over the 70-minute run.

7.2 Rate of drying curve



Critical free-moisture content, X_c	0.040 kg/kg dry solid
Rate of drying at critical point, R_c	0.0346 kg/(m ² .min)

7.3 Total time of drying

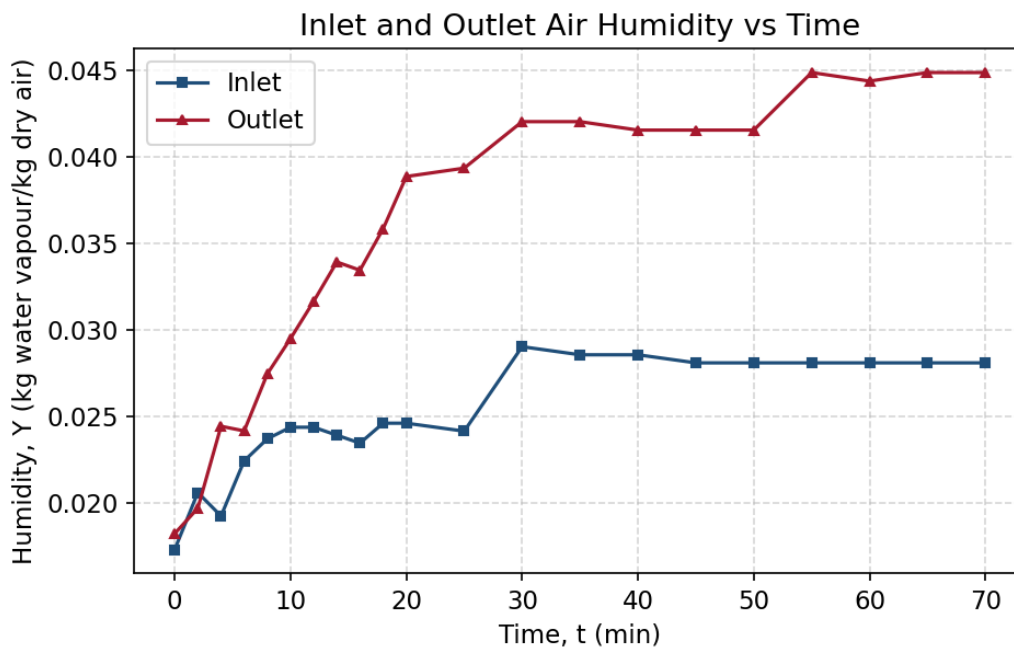
Using $X_1 = 0.220$, $X_c = 0.040$, and $X_2 = 0.008$ (the last independently measured point, T19 – see note below) in Eq. (1):

$$t_T = \frac{0.250}{0.0245 \times 0.0346} \left[(0.220 - 0.040) + 0.040 \ln \frac{0.040}{0.008} \right] = 72.0 \text{ min}$$

	Theoretical (Eq. 1)	Experimental
Total drying time	72.0 min	70 min

Note: $X_2 = 0$ exactly at T20, but this is tautological – T20’s own weight was used to define the dry-basis reference w_2 , so its X_T is zero by construction rather than by independent measurement. Since Eq. (1)’s $\ln(X_c/X_2)$ term diverges as $X_2 \rightarrow 0$, the last independently measured non-zero point (T19, $X_T = 0.008$) is used as the practical final free-moisture content.

7.4 Humidity of inlet and outlet air



Outlet humidity exceeds inlet humidity throughout the run (after the initial transient), consistent with the air picking up moisture evaporated from the wet solid as it passes through the dryer.

8 Precautions

1. Ensure the wet-bulb thermometer cotton wick remains fully wetted throughout the run for accurate wet-bulb readings.
2. Maintain a steady, constant air velocity throughout the run using the anemometer to avoid disturbing the drying rate.
3. Take weight readings promptly and consistently at the specified time intervals to minimise timing errors.
4. Ensure the pan is placed back in the drying chamber immediately after wetting the brick particles to minimise pre-drying moisture loss before $t = 0$.

9 References

1. Warren, L. McCabe, Smith, J. C., and Harriott, P., *Unit Operations of Chemical Engineering*, 6th edition, McGraw Hill, New Delhi, India, 2000.
2. Richardson, J. F., Harker, J. H. and Backhurst, J. R., *Coulson and Richardson's Chemical Engineering*, 5th edition, Vol-2, Asian Books Private Limited, New Delhi, India, 2002.